

Section 1

THE RESKILLING IMPERATIVE

1

RESKILLING IN THE AGE OF AI

by Jorge Tamayo, Leila Doumi, Sagar Goel, Orsolya Kovács-Ondrejko, and Raffaella Sadun

Back in 2019 the Organisation for Economic Co-operation and Development made a bold forecast. Within 15 to 20 years, it predicted, new automation technologies were likely to eliminate 14% of the world's jobs and radically transform another 32%. Those were sobering numbers, involving more than 1 billion people globally—and they didn't even factor in ChatGPT and the new wave of generative AI that has recently taken the market by storm.

Today advances in technology are changing the demand for skills at an accelerated pace. New technologies can not only handle a growing number of repetitive and manual tasks but also perform increasingly sophisticated kinds of knowledge-based work—such as research, coding, and writing—that have long been considered safe from disruption. The average half-life of skills is now less than five years, and in some tech fields it's as low as two and a half years. Not all knowledge workers will lose their jobs in the years ahead, of course, but as they carry out their daily tasks, many of them may well discover that AI and other new technologies have so significantly altered the nature of what they do that in effect they're working in completely new fields.

To cope with these disruptions, a number of organizations are already investing heavily in upskilling their workforces. One recent BCG study suggests that such investments represent as much as 1.5% of those organizations' total budgets.¹ But upskilling alone won't be enough. If the OECD estimates are correct, in the coming decades millions of workers may need to be entirely reskilled—a fundamental and profoundly complex societal challenge that will require workers not only to acquire new skills but to use them to change occupations.

Companies have a critical role to play in addressing this challenge, and it's in their best interests to get going on it in a serious way right now. Among those that have embraced the reskilling challenge, only a handful have done so effectively, and even *their* efforts have often been subscale and of limited impact, which leads to a question: Now that the need for a reskilling revolution is apparent, what must companies do to make it happen?

In our work at the Digital Data Design Institute at Harvard's Digital Reskilling Lab and the BCG Henderson Institute we have been studying this question in depth, and as part of that effort we interviewed leaders at almost 40 organizations around the world that are investing in large-scale reskilling programs. During those interviews we discussed common challenges, heard stories of early success, and discovered that many of those companies are thinking in important new ways about why,

when, and how to reskill. In synthesizing what we've learned, we've become aware of five paradigm shifts that are emerging in reskilling—shifts that companies will need to understand and embrace if they hope to succeed in adapting dynamically to the rapidly evolving era of automation and AI.

In this article we'll explore those shifts. We'll show how some companies are implementing them, and we'll review the unexpected challenges they've encountered and the promising wins they've achieved.

1. Reskilling Is a Strategic Imperative

During times of disruption, when many jobs are threatened, companies have often turned to reskilling to soften the blow of layoffs, assuage feelings of guilt about social responsibility, and create a positive PR narrative. But most of the companies we spoke with have moved beyond that narrow approach and now recognize reskilling as a strategic imperative. That shift reflects profound changes in the labor market, which is increasingly constrained by the aging of the working population, the emergence of new occupations, and an increasing need for employees to develop skills that are company-specific. Against this backdrop effective reskilling initiatives are critical, because they allow companies to build competitive advantage quickly by developing talent that is not readily available in the market and filling skills gaps that are instrumental to achieving their strategic objectives—before and better than their competitors do.

In recent years several major companies have embraced this approach. Infosys, for example, has reskilled more than 2,000 cybersecurity experts with various adjacent competencies and capability levels. Vodafone aims to draw from internal talent to fill 40% of its software developer needs. And Amazon, through its Machine Learning University, has enabled thousands of employees who initially had little experience in machine learning to become experts in the field.

Some companies now consider reskilling a core part of their employee value proposition and a strategic means of balancing workforce supply and demand. At those companies employees are encouraged to reskill for roles that appeal to them. Mahindra & Mahindra, Wipro, and Ericsson have policies, tools, and IT platforms that promote reskilling resources and available jobs—as does McDonald's, where restaurant employees have access to an app called Archways to Opportunity that maps skills learned on the job to career paths within the company and in other industries.

Finally, some companies are using reskilling to tap into broader talent pools and attract candidates who wouldn't otherwise be considered for open positions. ICICI Bank—headquartered in Mumbai and employing more than 130,000 people—runs an intense, academy-like reskilling program that prepares graduates, often from diverse backgrounds, for frontline managerial jobs. The program reskills some 2,500 to 4,000 employees each year. CVS used a similar approach during the Covid pandemic to hire, train, and onboard people (some of them laid-off hospitality workers) to create capacity for its critical vaccine and testing services.

2. Reskilling Is the Responsibility of Every Leader and Manager

Traditionally, reskilling is considered part of the overall corporate-learning function. When that's the case, responsibility for the design and implementation of the program is often siloed within HR, and its failure or success is measured very narrowly—in terms of the number of trainings delivered, the cost per learner, and similar training-specific metrics. According to a recent BCG report, only 24% of polled companies make a clear connection between corporate strategy and reskilling efforts. Reskilling

investments need a profound commitment from HR leaders, of course, but unless the rest of the organization understands the strategic relevance of those investments, it's very hard to obtain the relentless and distributed effort that such initiatives require to succeed.

At most of the organizations where we interviewed, reskilling initiatives are visibly championed by senior leaders, often CEOs and chief operating officers. They work hard to articulate for the rest of the company the connection between reskilling and strategy and to ensure that leadership and management teams understand their shared responsibility for implementing these programs. For example, as part of its ongoing digital transformation, Ericsson has developed a multiyear strategy devoted to upskilling and reskilling. The effort involves systematically defining critical skills connected to strategy, which correspond to a variety of accelerator programs, skill journeys, and skill-shifting targets—most of them dedicated to transforming telecommunications experts into AI and data-science experts. The company considers this a high-priority, high-investment project and has made it part of the objectives and key results that executives review quarterly. In just three years Ericsson has upskilled more than 15,000 employees in AI and automation.

Similarly, the executive team at CVS has made training and reskilling an integral part of the company's business strategies. Each individual business leader is now responsible for designing and delivering workforce-reskilling plans to help the company reach its goals, and the ability to do so is factored into performance assessments. Amazon, too, has famously committed to reskilling as a core strategic objective and now mentions it prominently in its leadership manifesto for managers. The visibility of this commitment contributes to Amazon's ability to achieve scale in reskilling programs.

3. Reskilling Is a Change-Management Initiative

To design and implement ambitious reskilling programs, companies must do a lot more than just train employees: They must create an organizational context conducive to success. To do that they need to ensure the right mindset and behaviors among employees and managers alike. From this perspective, reskilling is akin to a change-management initiative, because it requires a focus on many different tasks simultaneously.

Let's consider several of the most important.

Understanding supply and demand

To create a successful reskilling program, companies need a sophisticated understanding of supply (skills available internally and externally) and demand (skills needed to beat the competition). A useful way to develop this understanding is with a "skill taxonomy"—a detailed description of the capabilities needed for each occupation at a company. Employers used to put a lot of effort into creating such taxonomies from scratch, but many leading companies now rely on external providers for the bulk of the work. HSBC, for example, has adopted the taxonomy published by the World Economic Forum and customized it slightly to add skills specific to parts of its business. Similarly, SAP, which used to maintain an in-house taxonomy of 7,000 skills, has recently started working with Lightcast, which keeps a continually updated skill database. But developing a skill taxonomy is only the first step. Next comes the difficult job of deciding which skills get mapped to which jobs. Managers from different divisions may disagree about this. Such disagreement is often symptomatic of a deeper misalignment, and companies will need to resolve that before they undertake any major reskilling initiative.

Leaders must also determine what skills they will need in the future—a dynamic process that's critical for strategic reskilling programs. To do that well, they should focus on figuring out what skills the current strategy demands. Here they'll need to develop a rigorous strategic workforce-planning

methodology. The European insurance company Allianz has done interesting work on this front: It regularly translates forecasted business growth into talent demand, focusing on the number of people needed in various jobs and the skills they'll require. The model, which is updated as part of the annual planning process, involves economic scenario planning and takes into account the possible effects of digitalization on the workforce.

Recruiting and evaluating

Traditionally, candidates are recruited for training opportunities or internal roles on the basis of their degrees or relevant work experience, but that obviously doesn't work for reskilled workers. A well-developed skill taxonomy can help here, by allowing organizations to think about enrollment policies in light of skill adjacencies, which can facilitate the transition from one skill set to another. Novartis has implemented an AI-powered internal talent marketplace that predicts, matches, and offers roles and projects related to employees' skills and goals. In our research we've also found that if reskilling programs are to succeed, companies must develop a clear set of enrollment criteria for employees, not all of whom will have the right combination of motivation and personality traits to be a good fit for reskilling.

Shaping the mindset of middle managers

Middle managers are often resistant to the idea of reskilling, for two main reasons: They worry (1) that their reports won't be able to keep up with their regular responsibilities while being reskilled, and (2) that once their reports *are* reskilled, they'll move to other parts of the organization. In both cases this can lead to "talent hoarding," in which managers try to hold on to their favorite reports by denying them the ability to participate in reskilling. Several of the companies we spoke with have addressed this problem by making talent development an explicit managerial responsibility. Wipro evaluates managers according to their teams' participation in training offers, and Amazon promotes leaders on the basis of a performance assessment that includes the question "How have you developed your team?" Middle managers may also resist the idea of hiring reskilled employees, believing that they're not as desirable as traditionally skilled workers. This problem can be addressed by involving managers in the design and delivery of reskilling programs and by providing sensitivity and unconscious-bias training. No matter what form the resistance takes, senior leaders' role modeling in support of reskilling is vital to overcoming it.

Building skills into the flow of work

It can be costly and logistically challenging to take employees away from their day jobs to participate in training. And adults tend not to like or learn well in classroom-style situations. In a 2021 BCG survey 65% of the 209,000 participating workers said they prefer to learn on the job. As a result, the best approach for reskilling is to do as much training as possible by means of shadowing assignments, internal apprenticeships, and trial periods. The reskilling program at ICICI Bank, for example, consists of a four-month vocational residency, during which employees take part in simulation-style trainings for the managerial role they hope to get, and an eight-month deployment in the field that involves a structured internship in a bank branch and closely shadowing a current manager.

Matching and integrating reskilled employees

Employees need to be matched with new jobs. Our interview data shows that if destination roles are clearly described in advance, employees become more interested in reskilling because new career trajectories become apparent to them, and the reskilling itself becomes more effective because it's more position-specific. Once in their new jobs, reskilled employees need several kinds of support to integrate successfully: help with learning new work norms and culture, building networks, and developing soft skills. Here coaching and mentoring can be particularly effective tools. Amazon has demonstrated leadership in this area: It runs a variety of mentoring programs for reskilled employees, among them a buddy system, part of its Grow Our Own Talent program, that connects previous and current program participants. The company also provides career coaching for employees who are making particularly difficult transitions, such as from warehouse worker to software developer.

4. Employees *Want* to Reskill—When It Makes Sense

Many of the companies we spoke with mentioned that one of their biggest challenges was simply persuading employees to embark on reskilling programs. That's understandable: Reskilling requires a lot of effort and can set a major life change in motion, and the outcome isn't guaranteed. The OECD reports that only a very small fraction of workers typically take part in standard training programs, and those who do are often the ones who need them the least.

But workers may be more willing to engage in reskilling than prior data suggests. BCG data shows, for example, that 68% of workers are aware of coming disruptions in their fields and are willing to reskill to remain competitively employed. The key to success in this domain, our interviews suggest, is to treat workers respectfully and make the benefits of their participation in reskilling initiatives clear. As one of our interviewees explains, "The secret to scaling up reskilling programs is to design a product your employees actually like."

So how can organizations do that? We have several suggestions.

Treat employees as partners

Because reskilling programs are often associated with organizational disruption and job loss—or at least job change—leaders often avoid talking openly about the rationale for the programs and the opportunities they present. But employees are more likely to participate if they understand why the programs are being implemented and have had a role in creating them. Aware of this, several of the companies we spoke with made a point of being honest and clear about why they were creating reskilling programs and involving workers early. One large auto manufacturer, for example, told its diesel engineers that because of changes in the automobile industry, it had less and less need for their skills; it presented its program as a way of ensuring that they would have new jobs and job security in the years ahead. The companies also told us that in designing and implementing reskilling programs, it's critical to align with worker councils and unions early on and to involve them in advocating for the programs.

Design programs from the employee point of view

Reskilling programs require participants to make a major investment of time. So it's important to try to reduce the risk, cost, and effort involved and to provide (almost) guaranteed outcomes. Amazon allows employees in its Career Choice program to pursue everything from bachelor's degrees to certificates—and covers all costs in advance. That has proved to be a key factor in scaling up the program, which has already had more than 130,000 participants. CVS, for its part, uses an effective “train in place” model for new employees.

Dedicate adequate time and attention to the task

Because reskilling involves occupational change, it usually requires intensive learning, which is possible only if employees have the time and mental space they need to succeed. To that end, four times a year Vodafone dedicates days during which employees may devote themselves entirely to learning and personal development. Bosch goes even further: To help traditional engineers at the company earn degrees and get training in emerging fields, its Mission to Move program covers the cost of tuition and time spent learning for as much as two days a week for a whole year. It even gives participants days off before exams to prepare.

Naturally, providing employees the time and space for skilling can be harder in industries where most workers are hourly or shift-based. Iberdrola, a renewable energy company, faced this challenge as it digitized. Because it was embracing new technologies, the company realized it would need to reskill 3,300 employees in various hourly roles. Its leaders got the job done by working closely with frontline managers to ensure that operations weren't disrupted by workers' taking time off for training. The company considered all training hours to be work hours and paid employees for them accordingly.

5. Reskilling Takes a Village

Companies have tended to think of reskilling as an organization-level challenge, believing that they have to do the job by and for themselves. But many of the companies where we interviewed have recognized that reskilling takes place in an ecosystem in which a number of actors have roles to play. Governments can incentivize reskilling investments by means of funds, policies, and public programs; industry can team up with academia to develop new skill-building techniques; and NGOs can play a role in connecting corporate talent needs with disadvantaged and marginalized talent groups. Coalitions of companies may be more effective at the reskilling challenge than single organizations are.

When designing reskilling programs for the rapidly evolving era of AI and automation, companies need to harness the potential of this wider ecosystem. We've identified several ways in which they can do so.

Consider industry partnerships

Instead of thinking of themselves as competitors for a limited talent pool, companies can team up to conduct joint training efforts, which may significantly attenuate some of the challenges outlined above. For example, industry-wide skill taxonomies would provide a useful infrastructure and could in some cases help companies pool the knowledge and resources needed to invest in certain types of capabilities, such as cutting-edge AI skills, which are so new that individual organizations may not yet have the knowledge or the capacity to develop solutions on their own. Industry coalitions could also reassure participants that their investments in learning might open up broader future opportunities.

The Technology in Finance Immersion Programme, offered by the Institute of Banking and Finance Singapore, a nonprofit industry association, is a case in point. The program aims to build up an industry pipeline of capabilities in key technology areas, with participation from all major banks, insurance players, and asset managers in the country, to meet the talent needs of the financial services sector. Similarly, within the European Union a variety of stakeholders have formed the Automotive Skills Alliance, which is dedicated to the “re-skilling and up-skilling of workers in the automotive sector.”

Partner with nonprofits to reach diverse talent

Many reskilling nonprofits work with populations that are underrepresented in the workforce. By teaming up with these nonprofits, companies can significantly expand access to talent and employment opportunities in ways that benefit both parties, often at low cost. Some of the ongoing reskilling efforts we learned of in our research involve corporate partnerships with such innovative entities as OneTen (which helps Black workers in the United States), Year Up (which helps disadvantaged youths in the United States), Joblinge (which helps disadvantaged youths in Germany), and RISE 2.0 (a BCG program that helps workers in Singapore without a digital background move into digital roles). Year Up stands out among these initiatives for its careful use of statistical techniques to study the impact of its training on participants. Since 2011 the program has placed more than 40,000 young people in corporate roles and internships that would have been inaccessible to them without the reskilling support and network it provided. The program has an 80% placement rate at more than 250 participating companies.

Partner with local colleges and training providers

Companies have a lot to gain by teaming up with educational institutions in their reskilling efforts. Examples of such partnerships include the UK government-funded Institutes of Technology, which brings together colleges and major employers to provide practical technical training for workers without tech backgrounds in ways that allow companies to quickly react to new technologies and meet rapidly evolving skills needs; and BMW’s collaboration with the German Federal Employment Agency and the Association of German Chambers of Industry and Commerce, which supports the transition to electric vehicles with reskilling programs aimed at industrial electricians.

...

Many companies have an intuitive understanding of the need to embrace the reskilling paradigm shifts discussed in this article, and some, admirably, have already made tremendous commitments to doing so. But their efforts are hampered by two important limitations: a lack of rigor when it comes to the measurement and evaluation of what actually works and a lack of information about how to generalize and scale up the demonstrably successful features of reskilling programs. To adapt in the years ahead to the rapidly accelerating pace of technological change, companies will have to develop ways to learn—in a systematic, rigorous, experimental, and long-term way—from the many reskilling investments that are being made today. Only then will the reskilling revolution really take off.

Measuring and Evaluating Impact of Reskilling Initiatives

Informed by additional research we’ve conducted since publishing this article, we’ll add a sixth shift: the measurement and evaluation of the impact of these initiatives.

Any organization engaging in reskilling efforts needs to know whether those efforts are producing results. We recently surveyed chief human resource officers from approximately 1,200 organizations in the United States, along with business leaders from around 200. Although many organizations (86%) have reported running a pilot, that's only the first step to a more rigorous approach of experimenting with and measuring reskilling.

Almost all of the surveyed organizations reported that they track some KPIs relative to their reskilling program, with the majority reviewing those KPIs quarterly. Most focused on immediate training outputs, such as the number of participants who have completed the program (57%), change in skillset (56%), and total hours of training completed (51%). However, few organizations go beyond training specifics either to measure their programs' impact on business or HR-related outcomes (33%) or to compare the cost and benefits of alternative approaches (21%), such as comparing the net returns from reskilling to external hiring. While 42% report a positive return on investment (ROI), 56% either did not calculate or were unable to ascertain the ROI from their reskilling endeavors. This gap in accountability and measurement underscores a broader oversight in the strategic implementation of reskilling programs and represents an area ripe for recalibration. Without tracking all relevant KPIs and evaluating them at an appropriate frequency, it's difficult to optimize interventions or to know if reskilling investments have achieved their strategic objectives.

Adapted from "Designing a Successful Reskilling Program" on hbr.org, July 10, 2024 (reprint #H08AJR).

TAKEAWAYS

In the rapidly evolving era of automation and AI, millions of workers may need to be reskilled, forcing them to develop new competencies or find entirely new occupations. Companies play a critical role in meeting this challenge, but few have taken it seriously. To address this, organizations must embrace five paradigm shifts:

- # Reskilling is a strategic imperative. Effective initiatives allow companies to build competitive advantage quickly—before and better than their competitors do.
- # It is the responsibility of every leader and manager. Throughout the organization, leaders must understand and communicate the strategic relevance of reskilling investments.
- # It is a change-management initiative. Reskilling requires a focus on many different tasks simultaneously.
- # Employees *want* to reskill—when it makes sense. Success hinges on treating workers respectfully and clearly communicating the benefits of reskilling initiatives to them.
- # It takes a village. Companies should consider partnerships with nonprofits, colleges, training providers, or others in their industry.

NOTE

1. Sagar Goel and Orsolya Kovács-Ondrejko, "Your Strategy Is Only as Good as Your Skills," [bcg.com](http://www.bcg.com/publications/2023/your-strategy-is-only-as-good-as-your-skills), January 26, 2023, www.bcg.com/publications/2023/your-strategy-is-only-as-good-as-your-skills.

Adapted from an article in Harvard Business Review, September–October 2023 (product #R2305C).

2

ORGANIZATIONS NEED A DYNAMIC APPROACH TO TEACHING PEOPLE NEW SKILLS

by Sari Wilde, Alison Smith, and Sara Clark

As employees and organizations adapt to hybrid and remote work norms, emerging technologies, and general business disruptions, the skills needed to succeed in today's work environment are shifting rapidly. This poses a major challenge for organizations, particularly in today's war for talent. Organizations are struggling to find a way to keep up with changing skills, but they can't rely solely on hiring strategies to meet their needs. Instead, organizations must find or develop the skills they need within their existing workforces.

To better understand how organizations are managing their shifting skills needs, we surveyed 6,500 employees, as well as 75 HR leaders. Except where indicated, our findings come from these 2020 surveys.¹ Our analysis revealed that most organizations used one of two approaches to ensure employees have the skills the business needs when it needs them:

- **Reactive.** Unfortunately, many organizations find themselves taking a reactive approach, scrambling to build new skills as needs arise. One HR leader from a large manufacturing organization shared the challenge of trying to keep up with requests from the business to develop new skills: "When we put together a learning solution, the business has already moved on." In organizations with a reactive approach, employees apply only 54% of the new skills they learn after 12 months. These organizations are simply too slow to get the skills to employees at the times they're needed most.
- **Predictive.** On the other side, more than 50% of HR leaders think the solution is to get ahead by predicting the business' future skills needs. Without a crystal ball, attempts at predicting future skills are more likely to lead to misplaced investments in wasted training or outdated skills. Our research finds that trying to predict skills is worse than reacting: Employees apply only 37% of the new skills they learn at organizations with a predictive approach—significantly less than the reactive approach.

Rather than making big investments in predictive approaches that may not work or resorting to a reactive approach, our research reveals a third option: the dynamic approach. This strategy casts skills management as a dynamic exercise that embraces ambiguity, makes peace with imperfection, and frees up HR, managers, and employees to move fast in responding to the things they know and can anticipate. In our current work environment, where employees are constantly seeking increased transparency, personalization, and choice, the dynamic approach empowers them with more information so that they can make the right choices to build the skills they need to stay current in their desired roles. Employees at organizations that use a dynamic skills approach apply 75% of the new skills they learn—double the new skills application realized with the predictive approach.

Here are three steps organizations can take to adopt a dynamic approach to skilling and reskilling employees.

Identify Changing Skills Needs

Most organizations today rely on managers and leaders to identify skills needs and HR to implement solutions. When leaders recognize that employees are missing essential skills, the assumption is that HR can help them develop them. More specifically, over 75% of HR leaders report that leaders within the learning and development subfunction are primarily responsible for skills at their organizations. But leaders outside of HR don't always know the talent implications of business goals, which can result in identifying the wrong skills gaps or overlooking important needs.

To spot and close skills gaps as they arise, regularly bring together input from employees, leaders, and customers by facilitating a network of stakeholders who can report on the specific skills needs in their areas. Together, these skills-sensing networks are able to monitor changing needs and ensure employees are prepared.

For example, Lloyds Banking Group, a financial services company headquartered in London and a Gartner client, takes an iterative approach to mapping skills needs. HR facilitates collaboration among a network of skills stakeholders that includes business leaders, partners from each business unit, and HR specialists. This collaboration helps them quickly identify not only skills gaps, but also local and enterprise-wide actions to close them. The stakeholders meet regularly to review a skills dashboard that contains information about employees' existing skills and the talent interventions planned to meet the skills needs of the business. They check for progress against the agreed-on interventions and escalate any changes that are likely to impact the enterprise-wide skills strategy. Through this approach, Lloyds is able to make data-backed decisions that ensure local and enterprise-wide skills needs are met.

Jump-Start Skills Development

Many organizations respond to today's rapidly evolving skills needs by providing more formal training. Unfortunately, according to Gartner's 2018 Shifting Skills Survey of more than 7,000 employees worldwide, there is no significant relationship between the time employees spend in formal virtual or classroom training and the percentage of skills they use. While thoughtful, formal training still has its place, for many in-demand skills, it's just too slow. All too often, by the time the training is created and delivered, the need has changed.

These quickly evolving skills needs require new, faster solutions—what Gartner calls “skills accelerators.” Skills accelerators leverage existing resources and expertise to enable upskilling support that's “good enough” to meet skills needs in a timely way. Enacting a sufficient solution in time is better than implementing a perfected training solution too late. In practice, this can look like:

- **Identifying skills adjacencies:** Building shortcuts to in-demand skills by identifying adjacent, stepping-stone skills from skills employees already have
- **Training “skills disseminators” to coach peers:** Upskilling a select cohort of motivated and influential employees and then having them coach their peers on new skills as the need arises
- **Delivering learning to employees when they need it most:** Using data to identify and tailor learning delivery to the moments when skills needs arise in the business

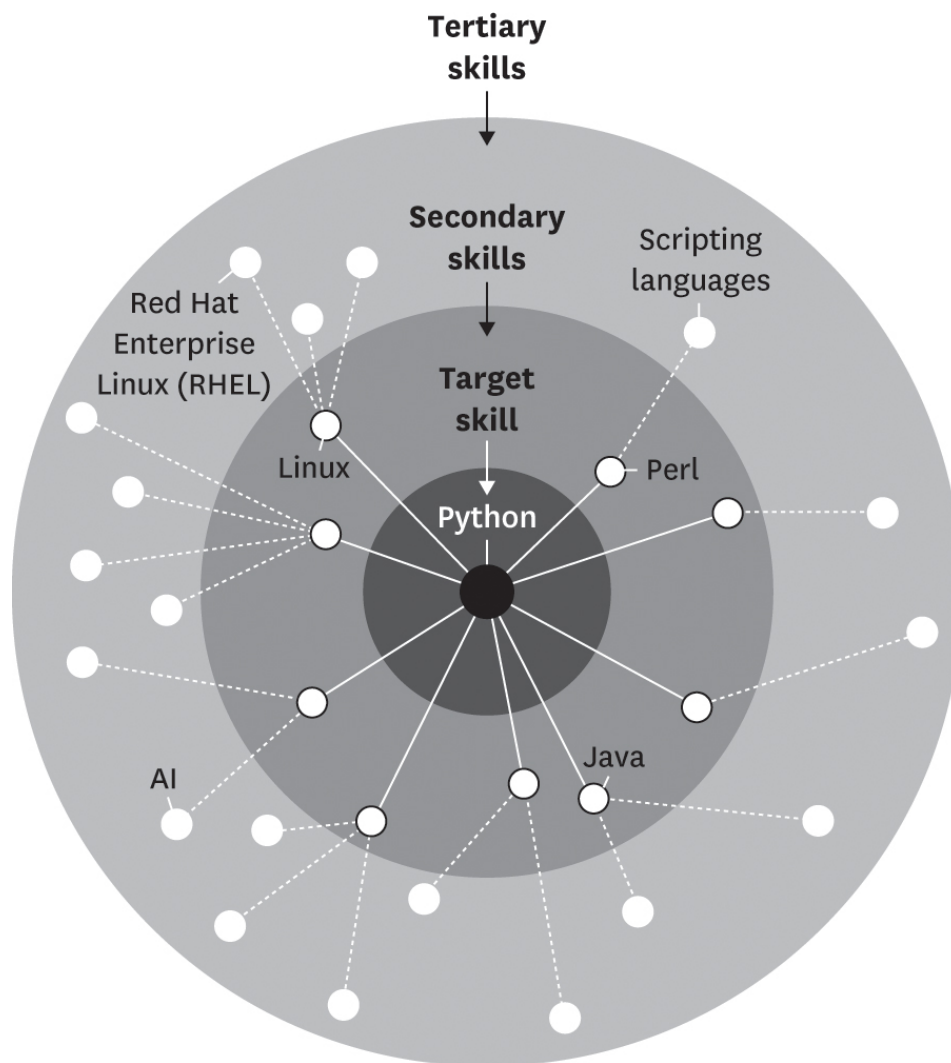
Identifying skills adjacencies can help business leaders tap into a broader and more diverse pool of employees and candidates who can get up to speed quickly. For instance, your organization may need an employee skilled in Python, a general-purpose programming language. Instead of limiting recruitment or internal searches to employees with knowledge of Python, a hiring manager should also consider candidates with closely related skills, such as Linux, Java, or Perl. Having one of these adjacent skills generally makes it easier for an employee to upskill—even through self-directed or on-the-job learning—in the desired area.

[Figure 2-1](#) shows the skills adjacencies for Python. Closely related skills like Java are part of an overall network of complementary skills that people with the Python skill usually have or can develop quickly. Tertiary skills, like other scripting languages, are related to Python but one more step removed.

FIGURE 2-1

Adjacent-skills map for the Python programming language

Linux, Java, and Perl skills are closely related to Python skills.



Source: Gartner TalentNeuron.

One large manufacturing organization we interviewed took this approach by thinking broadly about the backgrounds of current employees most likely to be able to develop data science skills. Instead of competing to hire experienced data scientists, they invested in developing these skills among employees with backgrounds in adjacent areas like mathematics, statistics, and business analysis.

Delivering learning to employees at the right times is another skills accelerator. Recognizing that time of learning and application were disconnected, one of Gartner's clients, CVS Health, a healthcare company that owns the CVS retail chain, focused on aligning learning delivery with moments of employee and business need. HR leaders realized that inputs from business leaders aren't always sufficient or timely enough to identify when new skills and learning are needed most. They use data from a variety of sources to identify:

- Moments in which employees need to apply new skills and can benefit most from learning (such as tenure milestones and promotions)
- Opportunities to build skills to support the business during high-demand moments (such as flu season)
- Real-time productivity or on-the-job performance data that HR accesses directly from the various business systems to trigger training, reskilling, or upskilling on a regular basis

Foster Transparency Between Employees and the Organization

Ensuring both the organization and employees are moving in the same direction is key to developing skills dynamically. While many leaders and managers try to motivate employees to engage in continuous learning by fostering a learning mindset, most employees are already motivated. In fact, 97% of employees report that they would learn a new skill if given the opportunity. Yet only 39% percent believe their organization is effective at helping them understand how information about skills needs applies to their own context.

To help employees make informed decisions about their development, leaders need to share evolving skills needs—even when plans are uncertain—and how these changes are likely to impact specific roles. Employees should also share their skills and career goals with the business. Exchanging this information empowers employees and leaders with the information they need to match and pursue mutually beneficial and flexible development opportunities.

“Connector managers”—those who can connect employees to the right people and resources at the right time—are particularly effective at diagnosing their employees’ strengths, development areas, motivations, and career aspirations. They’re also more transparent with employees about skill needs and opportunities.

In our interviews with HR leaders about their approaches to more transparently sharing skills information among leaders and employees, we found that a few leading organizations have started to ask employees to document skills as part of a portfolio or profile throughout their careers. At these organizations, employees use an HR-supplied portal to track current skills, knowledge, and experiences alongside their career goals and development aspirations. Leaders can also access this information to fill critical roles and direct employees to development opportunities based on their profiles. Helping employees understand the connection between tracking skills information and future career opportunities makes them more likely to commit to maintaining their profiles.

As industries, organizations, customer needs, and work norms continue to shift and evolve, the need for rapid reskilling and upskilling will only intensify. These challenges require organizations to rethink the boundaries of current solutions to skills gaps. Rather than trying to read a crystal ball to identify the skills of the future or waiting for requests from business leaders for new skills, taking a dynamic skills approach leads to the highest likelihood of employees actually applying the skills they learn in their current roles.

TAKEAWAYS

As industries, organizations, customer demands, and work norms continue to evolve, the need for rapid and responsive reskilling and upskilling intensifies. A dynamic skills approach increases the likelihood of employees applying new skills in their roles. There are three steps for adopting a dynamic approach to upskilling and reskilling employees:

- # **Identify changing skills needs.** Regularly gather input from employees, leaders, and customers through a network of stakeholders to monitor and report on specific skills needs.
- # **Jump-start skills development.** Identify skill adjacencies, train “skill disseminators” to coach peers, and provide learning to employees when they need it most.
- # **Foster transparency between employees and the organization.** Leaders must share evolving skills needs and potential impacts on roles with employees, helping their people make informed development decisions even when organizational plans are uncertain.

NOTES

1. “Gartner Survey Reveals HR Leaders’ Number One Priority in 2022 Will Be Building Critical Skills and Competencies,” press release, Gartner, October 20, 2021, www.gartner.com/en/newsroom/press-releases/2021-10-20-gartner-survey-reveals-hr-leaders--number-one-priorit.

Adapted from content posted on hbr.org, November 25, 2021 (product #H06PB9).